Shristi Bastola

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LinkedIn | GitHub

Education

Kathmandu University – School of Science — Dhulikhel, Kavre
B.Tech Environmental Engineering — Aug 2019 - May 2024
CGPA: 3.39/4

Courses Attended: Calculus and Linear Algebra, Numerical Methods, Statistics and Probability, Differential Equations & Complex Variables, Applied Hydrology, Meteorology &Climatology, GIS and Remote Sensing, Water and Wastewater Technology

Research Interests

- Machine learning and geospatial modeling for groundwater and hydrologic systems
- Climate–water interactions and data-driven environmental analysis

Research/Project Experience

Undergraduate project – Kathmandu University

1) Final year project: Suitability Analysis of Potential Landfill Site Selection Using Analytic Hierarchy Process (AHP) and GIS in Bharatpur Metropolitan City. (Feb 2024 - May 2024)

    - Geospatial Analysis & GIS: Used ArcMap 10.8 for spatial data processing, map creation, and suitability analysis.
    - Remote Sensing (RS): Applied satellite imagery and spatial datasets for Land use and Land cover (LULC) assessment .
    - Multi-Criteria Decision Making (MCDM): Implemented Analytic Hierarchy Process (AHP) and Weighted Linear Combination (WLC) for systematic decision-making.
    - Data Integration & Expert Consultation: Conducted virtual primary research by consulting municipality engineers and field professionals to identify key environmental, social, and economic criteria, and integrated their input with geospatial data for informed decision-making.

2) Physico-chemical Analysis of Quality of Drinking Water In Bhaisepati, Banepa (Jun 2022 - Nov 2022)
    - Conducted physico-chemical analysis of water samples and evaluated key parameters (pH, turbidity, TDS, EC, hardness, ions) using standard instrumental methods.
    - Collected field data and engaged with local communities to ensure accurate sampling and contextual understanding.
    - Assessed water quality by comparing results with national and WHO standards to determine suitability for human consumption.

Work Experience

Data Trainee (GIS and DRR Team) | NAXA Pvt. Ltd. — Oct 2024 - May 2025

- Collected, managed and processed household-level and infrastructure data using Google Earth Engine, Excel and GIS within standardised geodatabases.
- Collaborated with GIS and DRR teams to ensure data quality, integration, and consistency across projects.
- Conducted research on trekking routes in Nepal and performed geodata collection and validation for tourism-related projects.

- Coordinated and conducted QGIS-based digitization training sessions for a team of 15 digitizers supporting house-numbering projects across Birgunj Metropolitan City (32 wards, ~48,000 households) and Tokha Municipality, Kathmandu (11 wards, ~38,000 households).

Volunteering/Certificates

OMLAS, Nepal• OMLAS Champion Feb 2023 - Feb2024

- Participated in capacity-building workshops on project management, effective communication, and systems thinking.
- Gained practical experience in community-based projects and local leadership practices**.**

FECAM, Dhulikhel• Volunteer March 2022

- Supported the organization and logistics of an environmental engineering symposium involving students, faculty, and professionals.
- Assisted with event coordination, guest management, and session facilitation for speakers and participants.

FXB Climate Advocates Jan 2026 - May 2026

- Empowering Women Through Climate Literacy at the household level.

Training/Mentorship

Data Science and Machine Learning, Deerwalk Training Center Aug 2025 - Dec 2025

- Completed specialized training in machine learning and deep learning, covering neural networks, backpropagation, optimization algorithms, and predictive modeling workflows.
- Gained experience in designing, training, and evaluating data-driven models through feature engineering, model validation, and performance optimization.
- Applied machine learning techniques including regression, classification, clustering, and neural network-based methods using Python.
- Developed proficiency with data analytics and machine learning tools such as Pandas, NumPy, Scikit-learn, Matplotlib, and TensorFlow/Keras.

Asian Environmental Youth Network (AEYN), Mentee Sep 2025 - Nov 2025

- Land Use and Land Cover classification using GEE and Python

Annual Nepal AI School (ANAIS), Participant Dec 2025 - Jan 2026

- Winning team member of ideathon titled:- DEKHO: Real-Time Monocular Navigation System

Skills

GIS & Remote Sensing: QGIS, ArcGIS, Google Earth Engine

Programming & Data analysis Tools: Python**,** Excel, Google Sheets

Lab: Instrumental Analysis (UV-Vis spectrophotometer, pH, turbidity, and DO meters)

Microsoft office: Word & PowerPoint

English Proficiency: IELTS; 7